

## 3.3 Evaluation

Intrinsic versus extrinsic tests, and precision, recall, and F1.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

### 1. Two kinds of evaluation

A spam filter can be scored in two different rooms.

**Intrinsic** evaluation scores the NLP module on a held-out set with labels. Precision, recall, F1, accuracy. The AI team can compute these without waiting on the rest of the company.

**Extrinsic** evaluation scores the *product* outcome. How much time did people waste because spam landed in the inbox, or a real message landed in junk?

Intrinsic is cheaper and faster. It is a **proxy**. You run it first and often. You go to extrinsic only when intrinsic numbers are stably good. Bad intrinsic almost always means bad extrinsic. Good intrinsic does not guarantee a good product — that is why the second measurement exists.

Stakeholders outside the AI team usually own extrinsic numbers. That is another reason to keep intrinsic in-house and current.

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### 2. The confusion matrix

For a binary task, every test example falls in one cell.

|                   | Predicted positive  | Predicted negative  |
|-------------------|---------------------|---------------------|
| Actually positive | true positive (TP)  | false negative (FN) |
| Actually negative | false positive (FP) | true negative (TN)  |

Spam as the positive class: TP is junk you caught, FP is a real email you buried, FN is junk you missed.

| Metric                     | Description                                                                                                                                                                                                                                                         | Applications                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accuracy                   | Used when the output variable is categorical or discrete. It denotes the fraction of times the model makes correct predictions as compared to the total predictions it makes.                                                                                       | Mainly used in classification tasks, such as sentiment classification (multiclass), natural language inference (binary), paraphrase detection (binary), etc.                             |
| Precision                  | Shows how precise or exact the model's predictions are, i.e., given all the positive (the class we care about) cases, how many can the model classify correctly?                                                                                                    | Used in various classification tasks, especially in cases where mistakes in a positive class are more costly than mistakes in a negative class, e.g., disease predictions in healthcare. |
| Recall                     | Recall is complementary to precision. It captures how well the model can recall positive class, i.e., given all the positive predictions it makes, how many of them are indeed positive?                                                                            | Used in classification tasks, especially where retrieving positive results is more important, e.g., e-commerce search and other information-retrieval tasks.                             |
| F1 score                   | Combines precision and recall to give a single metric, which also captures the trade-off between precision and recall, i.e., completeness and exactness. F1 is defined as $(2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ . | Used simultaneously with accuracy in most of the classification tasks. It is also used in sequence-labeling tasks, such as entity extraction, retrieval-based questions answering, etc.  |
| AUC                        | Captures the count of positive predictions that are correct versus the count of positive predictions that are incorrect as we vary the threshold for prediction.                                                                                                    | Used to measure the quality of a model independent of the prediction threshold. It is used to find the optimal prediction threshold for a classification task.                           |
| MRR (mean reciprocal rank) | Used to evaluate the responses retrieved given their probability of correctness. It is the mean of the reciprocal of the ranks of the retrieved results.                                                                                                            | Used heavily in all information-retrieval tasks, including article search, e-commerce search, etc.                                                                                       |

| Metric                                 | Description                                                                                                                                                                                                                                         | Applications                                                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| MAP (mean average precision)           | Used in ranked retrieval results, like MRR. It calculates the mean precision across each retrieved result.                                                                                                                                          | Used in information-retrieval tasks.                                                                                                             |
| RMSE (root mean squared error)         | Captures a model's performance in a real-value prediction task. Calculates the square root of the mean of the squared errors for each data point.                                                                                                   | Used in conjunction with MAPE in the case of regression problems, from temperature prediction to stock market price prediction.                  |
| MAPE (mean absolute percentage error)  | Used when the output variable is a continuous variable. It is the average of absolute percentage error for each data point.                                                                                                                         | Used to test the performance of a regression model. It is often used in conjunction with RMSE.                                                   |
| BLEU (bilingual evaluation understudy) | Captures the amount of n-gram overlap between the output sentence and the reference ground truth sentence. It has many variants.                                                                                                                    | Mainly used in machine-translation tasks. Recently adapted to other text-generation tasks, such as paraphrase generation and text summarization. |
| METEOR                                 | A precision-based metric to measure the quality of text generated. It fixes some of the drawbacks of BLEU, such as exact word matching while calculating precision. METEOR allows synonyms and stemmed words to be matched with the reference word. | Mainly used in machine translation.                                                                                                              |
| ROUGE                                  | Another metric to compare quality of generated text with respect to a reference text. As opposed to BLEU, it measures recall.                                                                                                                       | Since it measures recall, it's mainly used for summarization tasks where it's important to evaluate how many words a model can recall.           |
| Perplexity                             | A probabilistic measure that captures how confused an NLP model is. It's derived from the cross-entropy in a next word prediction task.                                                                                                             | Used to evaluate language models. It can also be used in language-generation tasks, such as dialog generation.                                   |

### 3. Metrics you will compute

Let (P) be “predicted positive” and (T) be “actually positive.”

| Metric    | Formula                                                                       | Reads as                                        |
|-----------|-------------------------------------------------------------------------------|-------------------------------------------------|
| Accuracy  | $(\mathrm{TP} + \mathrm{TN}) / N$                                             | Overall fraction correct                        |
| Precision | $\mathrm{TP} / (\mathrm{TP} + \mathrm{FP})$                                   | Of what you flagged, how much was right         |
| Recall    | $\mathrm{TP} / (\mathrm{TP} + \mathrm{FN})$                                   | Of what was truly positive, how much you caught |
| F1        | $(2 \cdot \mathrm{prec} \cdot \mathrm{rec}) / (\mathrm{prec} + \mathrm{rec})$ | Harmonic mean; punishes a large gap             |

Accuracy is a poor headline when classes are unbalanced. A filter that never predicts spam is 98% accurate if 2% of mail is spam, and it is useless.

**Precision-heavy** tasks: you cannot afford false alarms (legal hold, medical alert). **Recall-heavy** tasks: you cannot afford misses (cancer mention, fraud). F1 is the compromise when both matter and you need one number.

For more than two classes, compute precision/recall/F1 **per class**, then macro-average (each class equal) or micro-average (each example equal). Say which one you report.

## 4. How to run it so it means something

- Hold out a **test** set you do not tune on. Use a validation split or cross-validation for decisions.
- Keep the same pre-processing on train and test (note **2.3**).
- Look at errors, not only the scalar. A confusion matrix and a few false positives/negatives tell you whether to change features or labels.
- If one country, language, or product line is systematically worse, that is a data problem (note **2.1**), not a reason to report only the average.

Lab 3 asks for accuracy, precision, recall, F1, **and** one extrinsic-style check (hard errors, a matrix, or a slice).

## 5. Beyond a single F1

A ranking or retrieval system needs precision@k or recall@k, not document-level F1. Generation (summary, translation) needs overlap or human ratings; that is later in the course.

For classification this week, also look at:

- **threshold:** moving it trades precision for recall. Plot both if the product can choose a cutoff.
- **calibration:** a score of 0.9 should be right about 90% of the time if someone will use the number as a probability.

- **slices:** F1 on the whole test set can hide a language or a country that is failing.

Those are still intrinsic. Extrinsic is “did the queue get shorter” or “did people open the email.” Do not call a prettier confusion matrix extrinsic.

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## 6. Practice

1. A spam model: TP 80, FP 20, FN 20, TN 880. Compute precision, recall, accuracy. Which number would you show a user who is angry about lost mail?
2. Why is F1 closer to the worse of precision and recall, not the average?
3. You raise recall on toxic-comment detection and precision collapses. What did you probably do to the threshold, and what extrinsic cost did you just buy?